92. Among which one of the following sets social/religious groups is the extent of poverty the highest, as per government statistics for the nineties?

(a) Muslims in Kerala, Gujarat and A.P.
(b) Tribals in Bihar, Orissa, M.P. and Maharashtra
(c) Scheduled Caste in Punjab, Western UP, Northern Rajasthan and Tamil Nadu
(d) Christians in Gujarat, and Assam

Ans. (b) : According to the recent data made available by Central Statistical Organisation, the incidence of poverty is highest in tribes of Bihar, Orissa and Madhya Pradesh.

93. Tourism industry in India is quite small compared to many other countries in terms of India's potential and size. Which one of the following statements is correct is this regard?

(a) Distances is India are too for apart and its luxury hotels are too expensive for western tourists
(b) For most of the months India is too hot for western tourists of feel comfortable
(c) Most of the picturesque resorts in India such as in the North-East and Kashmir are, for all practical purposes, out of bounds
(d) In India, the infrastructure required for attracting tourists is inadequate

Ans. (d) : India due to its handicrafts, culture and monuments is a preferred destination for tourism. However, there is a lack of tourist infrastructure.

94. The Employment Assurance Scheme envisages financial assistance to rural areas for guaranteeing employment to at least.

(a) 50 percent of the men and women seeking jobs in rural areas
(b) 50 percent of the men seeking jobs in rural areas
(c) One man and one woman in a rural family living below the poverty line
(d) One person in a rural landless household living below the poverty line

Ans. (c) : Employment Assurance Scheme was launched on 2nd October, 1993 for implementation in 1778 identified backward blocks of different States. Later, the scheme was extended to the remaining blocks of the country in a phased manner. At present, the scheme is being implemented in all the rural blocks of the country. The programme has been restructured from 1.4.99. The primary objective of the EAS is to provide gainful employment during the lean agricultural season to all able bodied adults in rural areas who are in need and desirous of work, but cannot find work. The secondary objective is the creation of community, social and economic assets for sustained employment and development. Internal debt comprises loans raised in the open markets.

PHYSICS

95. Barium in a suitable form is administered to patients before an X-ray examination of the stomach, because:

(a) Barium allows X-ray to pass through the stomach on account of its transparency to X-rays
(b) Barium compound, like magnesium sulphate helps in cleaning the stomach before X-ray examination
(c) Barium is a good absorber of X-rays and this helps the stomach to appear clearly in contrast with the other regions in the picture.
(d) Barium salts are white in colour and this helps the stomach to appear clearly in contrast with other regions in the picture.

Ans. (c) : Barium is given in adequate amount to patients before X-ray examination. The gastrointestinal tract does not show up very well on ordinary X-ray pictures. However, if a patient drinks a white liquid which contains a chemical called barium sulphate, the outline of the upper parts of the gut (esophagus, stomach and small intestines) shows up clearly on X ray pictures as X-rays do not pass through barium.

96. Endoscopy, a technique used to explore the stomach or other inner parts of the body is based on the phenomenon of:

(a) Total internal reflection
(b) Interference
(c) Diffraction
(d) Polarization

Ans. (a) : Endoscopy, a technique used to explore the stomach or other inner parts of the body is based on the principle of total internal reflection. The endoscopy procedure uses an endoscope to examine the interior of a hollow organ or cavity of the body. Unlike many other medical imaging techniques, endoscopes are inserted directly into the organ.

97. A hemispherical bowl is filled to the brim with a beverage. The contents of the bowl are transferred into a cylindrical vessel whose radius is 50% more than its height. If the diameter is same for both bowl and cylinder, then the volume of the beverage in the cylindrical vessel will be:

(a) $66\frac{2}{3}\%$
(b) $78\frac{1}{2}\%$
(c) 100%
(d) More than 100% (that is, some liquid will still be left in the bowl)

Ans. (c) : Let the radius of the hemispherical bowl be 'r'

$\therefore$ volume of the bowl $= \frac{2}{3}\pi r^3$

as the beverage is filled to the brain

$\therefore$ volume of the beverage $= \frac{2}{3}\pi r^3$ .....(i)

Now radius of the cylindrical vessel = r (given that diameter is same as the bowl).
and height of the vessel =

$$r = h + \frac{50}{1w}h$$

$$= h + \frac{1}{2}h \Rightarrow \frac{3}{2}h$$

$$h = \frac{2}{3}r$$

Now volume of the vessel $= \pi r^2 h$

$$= \pi r^2 \times \frac{2}{3}r$$

$$= \frac{2}{3}\pi r^3 \quad \text{.....(ii)}$$

as the volume is equal to the volume of beverage.
100% beverage can be transferred to the vessel.

98. A hunter aims his gun at a point between the eyebrows of a monkey sitting on a branch of a tree. Just as he fires, the monkey jumps down. The bullet will:

(a) Hit the monkey at the point aimed
(b) Hit the monkey below the point aimed
(c) Hit the monkey above the point aimed
(d) Miss the monkey altogether

Ans. (a) : As monkey and bullet both fall under the same gravitational force, so bullet will hit exactly at the same point where it has been aimed.

99. Cobalt-60 is commonly used in radiation therapy because it emits:

(a) Alpha rays (b) Beta rays
(c) Gamma rays (d) X-rays

Ans. (c) : The most common radioactive isotope of cobalt is cobalt-60. It is an important gamma-emitting radionuclide produced as a fission product. The gamma ray (photon) energy can be used in radiotherapy such as for the treatment of cancer, in food irradiation, or in industrial gauges or sensors. Cobalt-60 is used in the radiotherapy of cancer. Gamma rays penetrating power is vey great and can pass through several centimeters of Lead.

100. Consider the following statements:

An ordinary light bulb has a rather short life because the–

1. Filament wire is not uniform
2. Bulb cannot be evacuated completely.
3. Wires supporting the filament melt at thigh temperature.

Which of the above statements are correct?

(a) 1 and 3 (b) 2 and 3
(c) 1 and 2 (d) 1, 2 and 3

Ans. (d) : An electric light bulb uses a glowing wire filament heated by electrical resistance to generate light (a process known as thermal radiation). The 'bulb' is the glass enclosure which keeps the filament in a vacuum or low-pressure noble gas. Complete evacuation of bulb is not possible. A small quantity of gases left behind which expands on heating through filament. Moreover, the wires supporting the filament sometimes melt due to high heat generated as a result of high resistance of filament. Thus, these factors leads to short life span of ordinary bulb.

101. Consider the following statement regarding a motor car battery:

1. The voltage is usually 12V.
2. Electrolyte used in hydrochloric acid.
3. Electrodes are lead and copper.
4. Capacity is expressed in ampere-hour.

Which of the above statements are correct?

(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 4

Ans. (d) : In case of motor car battery or automotive batteries a nominal 12-volt potential difference is provided by connecting six galvanic cells in series. Capacity of these batteries is expressed in ampere-hour. Electrolyte used is a solution of about 35% sulphuric acid and 65% water and electrodes used are plates of lead and separate plates of lead dioxide.

102. Match List-I (Quantity) with List-II (Units) and select the correct answer using the codes given below the lists:

List-I	List-II
A. High speed	1. Mach
B. Wavelength	2. Angstrom
C. Pressure	3. Pascal
D. Energy	4. Joule

Codes:

	A	B	C	D
(a)	2	1	3	4
(b)	1	2	4	3
(c)	1	2	3	4
(d)	2	1	4	3

Ans. (c) : Mach number (Ma or M) is Commonly used to represent the speed of an object when it is travelling close to or above the speed of +sound. Pressure is the force per unit area applied in a direction perpendicular to the surface of an object. The SI unit for pressure is the Pascal (Pa), equal to one Newton per square meter